

Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Test Report

Abstract and Executive Outline

Keywords: Prompt Caching; A/B Testing; Provider Routing; Cache Affinity; Latency; Throughput; Cost Attribution; mimo-v2.5

Abstract

This report evaluates `xiaomi/mimo-v2.5` on Infron and OpenRouter across cache reuse, observed cost, throughput, E2E latency, and Streaming TTFT under Prompt Caching workloads.

The main findings are: Infron leads Cache hit rate in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads Observed cost in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads Throughput in 3/4 routing modes; OpenRouter leads in 1/4; 0/4 tie; Infron leads E2E latency in 3/4 routing modes; OpenRouter leads in 1/4; 0/4 tie; Infron leads Streaming TTFT in 3/4 routing modes; OpenRouter leads in 1/4; 0/4 tie.

Overall, Infron's cross-mode strengths are throughput, Streaming TTFT, E2E latency, while OpenRouter's cross-mode strengths are observed cost, cache reuse. Platform choice should be driven by workload objective rather than a single headline metric.

Figure 0: Normalized Capability Radar

The five radar axes represent throughput, price, E2E latency, Streaming TTFT, and cache hit rate. Every metric is normalized to a 0–100 score, with farther outward meaning better.

Thick solid lines show platform–level contours; translucent lines and points show individual routing modes.

Conclusion Overview: Core Metric Winners by Routing Mode

Based on strict A/B paired samples. Blue represents Infron, orange represents OpenRouter; gold cells mark the winner for the routing objective.

Throughput Infron wins 3/4 Max advantage 72.59%, higher is better	Observed cost OpenRouter wins 3/4 Max advantage 10.84%, lower is better	E2E latency Infron wins 3/4 Max advantage 34.56%, lower is better	Streaming TTFT Infron wins 3/4 Max advantage 33.92%, lower is better	Cache hit rate OpenRouter wins 3/4 Max advantage 8.49%, higher is better
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Routing mode	Throughput objective	Cost objective	Latency objective	TTFT objective	Cache result
Throughput First throughput	Infron advantage 72.59%	OpenRouter advantage 31.56%	Infron advantage 15.89%	Infron advantage 30.65%	OpenRouter advantage 8.49%
Price First price	OpenRouter advantage 67.56%	Infron advantage 10.84%	OpenRouter advantage 34.56%	OpenRouter advantage 33.92%	Infron advantage 0.75%
Latency First latency	Infron advantage 6.04%	OpenRouter advantage 65.45%	Infron advantage 6.04%	Infron advantage 10.81%	OpenRouter advantage 0.04%
TTFT First ttft	Infron advantage 5.82%	OpenRouter advantage 27.99%	Infron advantage 5.81%	Infron advantage 10.71%	OpenRouter advantage 0.03%

Executive Outline

Dimension	Conclusion	Evidence
Controls	First/second <code>usage.prompt_tokens</code> deltas are limited to 50 tokens within each <code>sort/group/round</code> pair.	Methods and data quality
Cache reuse	Infron leads Cache hit rate in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie	Overall metrics and mechanism section
Observed cost	Infron leads Observed cost in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie	Overall metrics and provider drill-down
Performance	Infron leads Throughput in 3/4 routing modes; OpenRouter leads in 1/4; 0/4 tie; Infron leads E2E latency in 3/4 routing modes; OpenRouter leads in 1/4; 0/4 tie; Infron leads Streaming TTFT in 3/4 routing modes; OpenRouter leads in 1/4; 0/4 tie	Charts and statistical tests
Attribution boundary	Claims use observable response telemetry: usage, cost, TTFT, latency, provider fields, and cache tokens.	Provider/Route drill-down
Business meaning	Long-context, RAG-prefix, agent-tool, and high-frequency template workloads should evaluate cache rate, cost, first-token latency, and E2E latency together.	Discussion and conclusion

Routing–Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	OpenRouter	OpenRouter	Infron	Infron	Infron	Infron leads overall (3/5 metrics)
Price First	Minimize request and token cost	Infron	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)
Latency First	Minimize full–response waiting time	OpenRouter	OpenRouter	Infron	Infron	Infron	Infron leads overall (3/5 metrics)
TTFT First	Minimize streaming first–token time	OpenRouter	OpenRouter	Infron	Infron	Infron	Infron leads overall (3/5 metrics)

1. Introduction: Background, Questions, and Contributions

LLM inference–platform behavior is shaped by provider routing, prompt caching, streaming response handling, cost attribution, and fallback policy. This report treats the platform as an observable system and measures speed, cost, cache reuse, and first–token experience through strict A/B pairs.

1.1 Research Hypotheses

Hypothesis	Statement	Validation metric
H1	Stronger provider/cache affinity improves token–level cache hit rate for repeated stable prefixes.	Second–request cache–read tokens and token cache hit rate
H2	Higher cache hit rate can reduce observed cost, but does not necessarily reduce TTFT or E2E latency.	Observed cost, average TTFT, average request latency
H3	Different routing sorts change provider selection and produce different cost/throughput/latency frontiers.	Provider distribution, throughput, latency, cost

1.2 Contributions

- Uses response–returned `usage.prompt_tokens` as the input–token control while allowing small cross–platform accounting variance up to 50 tokens.
- Extends prompt–caching evaluation to cost, throughput, E2E latency, TTFT, provider distribution, reasoning telemetry, and paired statistical tests.
- Scopes every claim to observable response telemetry instead of private routing internals.

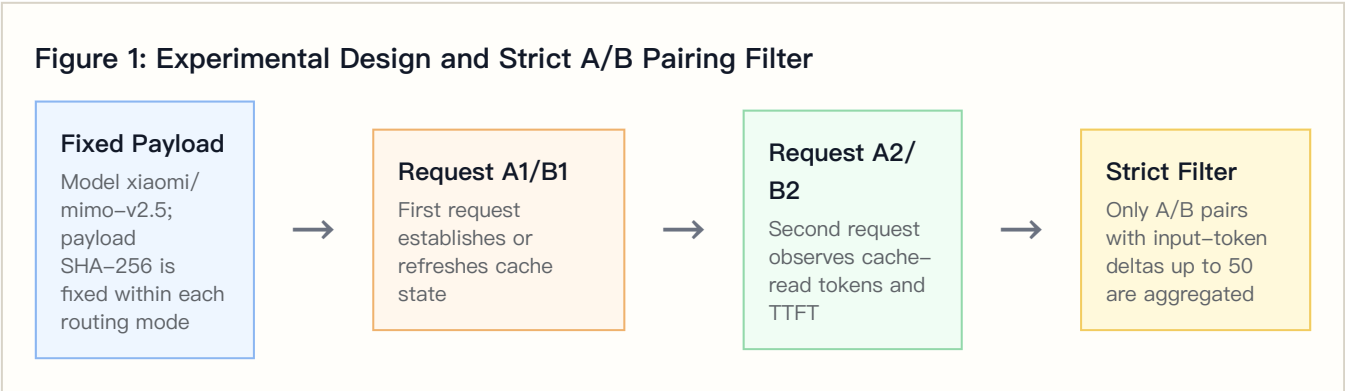
2. Experimental Design, Dataset, and Controls

2.1 Dataset Construction

The dataset is `business_representative`, covering 4 routing sorts, 2 platforms, 4 groups, and 50 rounds per group. Each round sends identical first/second Chat Completions requests; the second request observes cache-read tokens, TTFT, and E2E latency.

The built-in business templates represent stable long-context workloads such as RAG support, agent tool instructions, marketing automation, and code review.

2.2 Controlled Variables



Controlled-variable rule: within the same `sort/group/round`, both platforms must have first/second `usage.prompt_tokens` deltas no greater than 50 tokens. Total Input Tokens are computed from response-returned usage, not local tokenizer estimates.

2.3 Metric Definitions

Metric	Definition	Direction
Call cache hit rate	Share of second requests with <code>cache_read_tokens > 0</code>	Higher is better
Token cache hit rate	Second-request cache-read tokens / second-request prompt tokens	Higher is better
Observed cost	Sum of first + second request usage/cost values	Lower is better
Throughput	Completion tokens / request latency seconds	Higher is better
E2E latency	Full response latency per request	Lower is better
TTFT	Streaming first-token arrival time	Lower is better
Reasoning telemetry	Reasoning tokens from response usage	Used to explain latency, throughput, and cost

3. Experimental Environment and Data Quality

Item	Configuration
Model	xiaomi/mimo-v2.5
Provider model IDs	infron: xiaomi/mimo-v2.5 ; openrouter: xiaomi/mimo-v2.5
Platforms	Infron and OpenRouter
API protocol	/v1/chat/completions
Routing modes	Throughput First, Price First, Latency First, TTFT First
Groups	4
Rounds per group	50
Workers	24
Request mode	Streaming Chat Completions with TTFT collection
Reasoning / thinking control	No explicit reasoning/thinking parameter; model and platform defaults are preserved
Prompt length tiers	short ≈1500, medium ≈8000, long ≈32000
Excluded records	0

4. Results: Overall Metrics and Main Findings

Routing mode	Platform	Strict pairs	Total Input Tokens	Token cache hit rate	Observed cost	Throughput	E2E latency	Streaming TTFT
Throughput First	Infron	200	7075538	90.89%	\$0.53792900	3.90 tok/s	6103.12 ms	5073.92 ms
Throughput First	OpenRouter	200	7075538	98.61%	\$0.40888852	2.26 tok/s	7073.03 ms	6629.10 ms
Price First	Infron	200	7076338	99.88%	\$0.03393900	1.53 tok/s	8371.81 ms	7772.99 ms
Price First	OpenRouter	200	7075538	99.14%	\$0.40176022	2.57 tok/s	6221.51 ms	5804.38 ms
Latency First	Infron	200	7075538	99.73%	\$0.61243900	4.74 tok/s	3373.85 ms	2904.79 ms
Latency First	OpenRouter	200	7075538	99.77%	\$0.37016124	4.47 tok/s	3577.79 ms	3218.88 ms
TTFT First	Infron	200	7075538	99.73%	\$0.58492400	4.82 tok/s	3322.36 ms	2814.00 ms
TTFT First	OpenRouter	200	7075538	99.76%	\$0.45699362	4.55 tok/s	3515.33 ms	3115.36 ms

4.1 Tail Latency and Statistical Tests

Tail percentiles expose risk that averages hide.

Routing mode	Platform	P50 latency	P95 latency	P99 latency	P50 TTFT	P95 TTFT	P99 TTFT
Throughput First	Infron	4809.39 ms	13119.45 ms	15621.07 ms	4086.84 ms	11214.15 ms	14226.33 ms
Throughput First	OpenRouter	6115.97 ms	14679.31 ms	24490.92 ms	5788.92 ms	14550.51 ms	23248.41 ms
Price First	Infron	7638.25 ms	15210.05 ms	26151.63 ms	6948.13 ms	13991.88 ms	25962.27 ms
Price First	OpenRouter	5531.73 ms	12344.42 ms	23327.27 ms	5185.62 ms	11984.98 ms	22268.41 ms
Latency First	Infron	3076.35 ms	6085.01 ms	8006.44 ms	2601.66 ms	5494.32 ms	7328.23 ms
Latency First	OpenRouter	2992.94 ms	6003.49 ms	18724.96 ms	2574.46 ms	5684.02 ms	18634.10 ms
TTFT First	Infron	3027.00 ms	5484.81 ms	8204.21 ms	2543.74 ms	4957.15 ms	6726.54 ms
TTFT First	OpenRouter	3045.91 ms	6016.91 ms	14155.20 ms	2585.81 ms	5397.24 ms	13736.25 ms

Mean deltas use bootstrap 95% CIs; p-values use paired sign-flip permutation tests.

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Throughput First	Latency: OpenRouter – Infron	1939.81 ms	790.50 ms to 3123.88 ms	0.0018	200	Positive means lower Infron latency
Throughput First	TTFT: OpenRouter – Infron	3110.36 ms	2090.16 ms to 4230.80 ms	<0.001	200	Positive means lower Infron TTFT
Throughput First	Throughput: Infron – OpenRouter	1.6810 tok/s	1.3104 tok/s to 2.0173 tok/s	<0.001	200	Positive means higher Infron throughput
Throughput First	Cost: OpenRouter – Infron	\$-0.00064520	\$-0.00110266 to \$-0.00020186	0.0035	200	Positive means lower Infron cost
Throughput First	Token Cache Hit: Infron – OpenRouter	-10.74 pp	-15.20 pp to -6.34 pp	<0.001	200	Positive means higher Infron cache hit
Price First	Latency: OpenRouter – Infron	-4300.60 ms	-5433.80 ms to -3135.26 ms	<0.001	200	Positive means lower Infron latency
Price First	TTFT: OpenRouter – Infron	-3937.23 ms	-5075.90 ms to -2763.74 ms	<0.001	200	Positive means lower Infron TTFT
Price First	Throughput: Infron – OpenRouter	-1.4664 tok/s	-1.7029 tok/s to -1.2328 tok/s	<0.001	200	Positive means higher Infron throughput
Price First	Cost: OpenRouter – Infron	\$0.00183911	\$0.00147397 to \$0.00221194	<0.001	200	Positive means lower Infron cost
Price First	Token Cache Hit: Infron – OpenRouter	1.84 pp	0.36 pp to 3.82 pp	<0.001	200	Positive means higher Infron cache hit
Latency First	Latency: OpenRouter – Infron	407.88 ms	-305.93 ms to 1281.81 ms	0.3199	200	Positive means lower Infron latency
Latency First	TTFT: OpenRouter – Infron	628.17 ms	-70.44 ms to 1513.35 ms	0.1017	200	Positive means lower Infron TTFT

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Latency First	Throughput: Infron – OpenRouter	–0.1162 tok/s	–0.4115 tok/s to 0.1770 tok/s	0.4379	200	Positive means higher Infron throughput
Latency First	Cost: OpenRouter – Infron	\$–0.00121139	\$–0.00163065 to \$–0.00083766	<0.001	200	Positive means lower Infron cost
Latency First	Token Cache Hit: Infron – OpenRouter	–0.14 pp	–0.20 pp to –0.08 pp	<0.001	200	Positive means higher Infron cache hit
TTFT First	Latency: OpenRouter – Infron	385.93 ms	–265.20 ms to 1085.04 ms	0.2722	200	Positive means lower Infron latency
TTFT First	TTFT: OpenRouter – Infron	602.72 ms	–56.39 ms to 1273.67 ms	0.0722	200	Positive means lower Infron TTFT
TTFT First	Throughput: Infron – OpenRouter	0.1326 tok/s	–0.1410 tok/s to 0.4206 tok/s	0.3569	200	Positive means higher Infron throughput
TTFT First	Cost: OpenRouter – Infron	\$–0.00063965	\$–0.00090302 to \$–0.00040665	<0.001	200	Positive means lower Infron cost
TTFT First	Token Cache Hit: Infron – OpenRouter	–0.13 pp	–0.19 pp to –0.08 pp	<0.001	200	Positive means higher Infron cache hit

4.2 Reasoning / Thinking Control Check

This run does not explicitly set reasoning/thinking parameters and keeps model/platform defaults; this table records reasoning telemetry under default behavior.

Routing mode	Platform	Reasoning tokens	Avg reasoning tokens/request	Reasoning requests	Avg first reasoning token	Avg TTFT	Avg E2E latency
Throughput First	Infron	4285	10.7125	198	5178.47 ms	5073.92 ms	6103.12 ms
Throughput First	OpenRouter	7528	18.8200	400	6629.10 ms	6629.10 ms	7073.03 ms
Price First	Infron	0	0.0000	0	0.00 ms	7772.99 ms	8371.81 ms
Price First	OpenRouter	7671	19.1775	400	5804.38 ms	5804.38 ms	6221.51 ms
Latency First	Infron	0	0.0000	0	3011.02 ms	2904.79 ms	3373.85 ms
Latency First	OpenRouter	7661	19.1525	400	3218.88 ms	3218.88 ms	3577.79 ms
TTFT First	Infron	0	0.0000	0	2957.50 ms	2814.00 ms	3322.36 ms
TTFT First	OpenRouter	7746	19.3650	400	3115.36 ms	3115.36 ms	3515.33 ms

4.3 API Protocol Compatibility Matrix

This run uses `/v1/chat/completions` ; this table records success response, usage, cost, and cache telemetry coverage for both platforms under that protocol.

API protocol	Endpoint	Platform	Pairs	Requests	Success rate	Usage coverage	Token usage coverage	Cost coverage	Cache telemetry coverage
chat_completions	/v1/chat/completions	Infron	800	1600	100.00%	100.00%	100.00%	100.00%	100.00%
chat_completions	/v1/chat/completions	OpenRouter	800	1600	100.00%	100.00%	100.00%	100.00%	100.00%

5. Result Visualizations by Routing Mode

The following charts are driven by the strict-paired summary data and present routing-mode, platform, cost, cache, E2E latency, Streaming TTFT, and upstream provider differences.

5.1 Core Metric Chart Overview

The charts below are generated from the same strict-paired summary data.

**Figure 3-E2:
Normalized
capability
contour**

Five metrics are normalized to 0-100, where higher is better.

**Figure 3-E3:
Cost-cache
efficiency plane**

X-axis is token cache hit rate; Y-axis is observed total cost.

**Figure
3-
E1:
Routing-
mode
core
metric
comparison**

Infron and OpenRouter are shown side by side by routing mode.

Figure 3–E4: E2E latency and Streaming TTFT

Solid lines show E2E latency; dashed lines show Streaming TTFT.

Figure 3–E5: Upstream provider distribution

Provider shares explain cache–domain and performance differences.

Routing–Mode Drill–Down Charts

The horizontal axis shows relative advantage: blue to the right indicates Infron advantage, orange to the left indicates OpenRouter advantage.

Figure 4–E1: Throughput First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E2: Price First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E3: Latency First core metric comparison

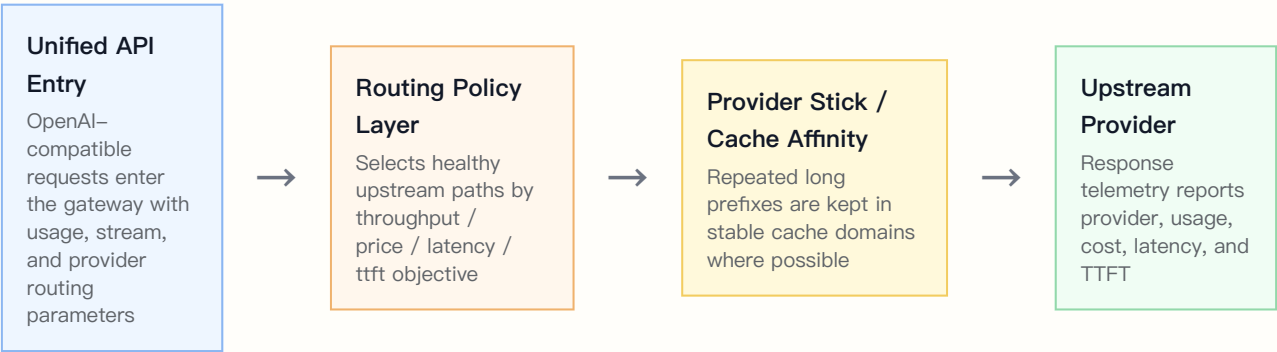
Direction and magnitude of relative advantage across five metrics.

Figure 4–E4: TTFT First core metric comparison

Direction and magnitude of relative advantage across five metrics.

6. Infron Technical Architecture and Cache/Cost Mechanism

Figure 12: Infron Multi-Provider Routing and Cache Control Plane



6.1 Multi-Provider Routing and Observable Control Plane

Requests enter a unified API, and the routing layer selects healthy upstream paths according to throughput, price, latency, or ttft objectives. Whether stable long prefixes stay in the same cache domain directly affects second-request cache-read tokens.

6.2 Provider Stick and Cache Affinity

Provider stick is cache affinity, not a permanent provider lock. Its goal is to reduce cache-domain fragmentation while respecting health and SLA constraints.

6.3 Cost-Control Path

Mechanism	Cache-rate effect	Cost effect	Observable signal
Stable prefix detection	Repeated prefixes are more likely to hit cache	Reduces repeated prefill cost	Payload SHA-256 and second-request cache-read tokens
Provider stick / cache affinity	Reduces cross-domain cache fragmentation	Avoids repeated cache warm-up	Provider distribution and token cache hit rate
Health checks and fallback	Protects availability while sometimes sacrificing cache	Reduces failure cost	HTTP status, provider distribution, tail latency
Cost-aware routing	Prefers lower-cost paths under constraints	Reduces total and per-round cost	Observed cost, cost breakdown coverage, cache-read tokens

7. Provider/Route Drill-Down

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	Infron	400	400	deepinfra 202, digitalocean 198

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	OpenRouter	400	400	DeepInfra 271, Xiaomi 129
Price First	Infron	400	400	xiaomi 400
Price First	OpenRouter	400	400	DeepInfra 256, Xiaomi 127, DigitalOcean 17
Latency First	Infron	400	400	deepinfra 400
Latency First	OpenRouter	400	400	DeepInfra 273, Xiaomi 127
TTFT First	Infron	400	400	deepinfra 400
TTFT First	OpenRouter	400	400	DeepInfra 299, Xiaomi 100, DigitalOcean 1

Upstream Provider Detail Distribution

Routing mode	Platform	Upstream provider	Requests	Share	first/second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens
Throughput First	Infron	deepinfra	202	50.50%	97/105	105	3863.66 ms	4453.84 ms	3720661	3232
Throughput First	Infron	digitalocean	198	49.50%	103/95	103	6308.64 ms	7785.72 ms	3354877	6298
Throughput First	OpenRouter	DeepInfra	271	67.75%	134/137	137	6599.68 ms	7062.81 ms	4315658	4336
Throughput First	OpenRouter	Xiaomi	129	32.25%	66/63	66	6690.91 ms	7094.49 ms	2759880	2064
Price First	Infron	xiaomi	400	100.00%	200/200	200	7772.99 ms	8371.81 ms	7076338	5140
Price First	OpenRouter	DeepInfra	256	64.00%	128/128	128	5336.89 ms	5761.04 ms	4152578	4096
Price First	OpenRouter	Xiaomi	127	31.75%	64/63	64	6897.18 ms	7235.06 ms	2755892	2032
Price First	OpenRouter	DigitalOcean	17	4.25%	8/9	9	4680.37 ms	5583.73 ms	167068	272
Latency First	Infron	deepinfra	400	100.00%	200/200	200	2904.79 ms	3373.85 ms	7075538	6400
Latency First	OpenRouter	DeepInfra	273	68.25%	136/137	137	2905.41 ms	3255.48 ms	4358746	4368
Latency First	OpenRouter	Xiaomi	127	31.75%	64/63	64	3892.71 ms	4270.63 ms	2716792	2032
TTFT First	Infron	deepinfra	400	100.00%	200/200	200	2814.00 ms	3322.36 ms	7075538	6400

Routing mode	Platform	Upstream provider	Requests	Share	first/second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens
TTFT First	OpenRouter	DeepInfra	299	74.75%	148/151	151	2931.66 ms	3327.89 ms	5466743	4784
TTFT First	OpenRouter	Xiaomi	100	25.00%	51/49	51	3542.48 ms	3953.99 ms	1606808	1600
TTFT First	OpenRouter	DigitalOcean	1	0.25%	1/0	1	15329.68 ms	15692.50 ms	1987	16

7.1 Cache–Rate and Cost Divergence Drill–Down

This table combines cache, cost, provider distribution, and reasoning telemetry to explain routing–mode differences.

Routing mode	Cache–hit delta	Infron cost multiple	Infron top path	OpenRouter top path	Reasoning token delta	Main attribution
Throughput First	–7.72 pp	1.32x	deepinfra 50.50%	DeepInfra 67.75%	–3243	OpenRouter has higher cache and lower cost; provider/ cache–domain mix is the main signal
Price First	+0.74 pp	0.08x	xiaomi 100.00%	DeepInfra 64.00%	–7671	Infron leads both cache and cost
Latency First	–0.04 pp	1.65x	deepinfra 100.00%	DeepInfra 68.25%	–7661	OpenRouter has higher cache and lower cost; provider/ cache–domain mix is the main signal
TTFT First	–0.03 pp	1.28x	deepinfra 100.00%	DeepInfra 74.75%	–7746	OpenRouter has higher cache and lower cost; provider/ cache–domain mix is the main signal

8. Stratified Results: Prompt–Length Cache Performance

This section aggregates second–request cache–read tokens, token–level cache hit rate, observed cost, E2E latency, and Streaming TTFT by prompt–length tier. Bold cells mark the advantaged side within each tier.

Prompt–Length Tier Overview

Prompt length tier	Target tokens	Platform	Pairs	Second prompt tokens	Second cache read tokens	Token cache hit rate	Observed cost	Avg E2E latency	Avg TTFT
short	1500	Infron	268	533178	506112	94.92%	\$0.08121300	4095.19 ms	3458.13 ms

Prompt length tier	Target tokens	Platform	Pairs	Second prompt tokens	Second cache read tokens	Token cache hit rate	Observed cost	Avg E2E latency	Avg TTFT
short	1500	OpenRouter	268	533044	521152	97.77%	\$0.06277326	3846.68 ms	3412.17 ms
medium	8000	Infron	268	2766974	2663488	96.26%	\$0.34829300	5183.66 ms	4536.25 ms
medium	8000	OpenRouter	268	2766840	2732544	98.76%	\$0.47585390	4706.89 ms	4295.43 ms
long	32000	Infron	264	10851324	10636416	98.02%	\$1.33972500	6619.30 ms	5949.43 ms
long	32000	OpenRouter	264	10851192	10801280	99.54%	\$1.09917644	6762.02 ms	6393.59 ms

Prompt Length x Routing Mode Cache Hit Rate

Prompt length tier	Routing mode	Infron	OpenRouter	Winner
short	Throughput First	85.29%	98.14%	OpenRouter
short	Price First	98.88%	96.65%	Infron
short	Latency First	97.76%	98.14%	OpenRouter
short	TTFT First	97.76%	98.14%	OpenRouter
medium	Throughput First	86.26%	99.50%	OpenRouter
medium	Price First	99.79%	96.55%	Infron
medium	Latency First	99.50%	99.50%	tie
medium	TTFT First	99.50%	99.50%	tie
long	Throughput First	92.35%	98.41%	OpenRouter
long	Price First	99.96%	99.92%	Infron
long	Latency First	99.89%	99.92%	OpenRouter
long	TTFT First	99.89%	99.91%	OpenRouter

9. Stratified Results: Group–Level Stability

Throughput First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	73.43%	\$0.11088400	14729.23 ms	13339.81 ms
Infron	2	50	50	91.74%	\$0.08954100	11593.76 ms	10912.12 ms
Infron	3	50	50	98.38%	\$0.19407000	6613.61 ms	5812.01 ms

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	4	50	50	99.72%	\$0.14343400	9392.97 ms	9110.71 ms
OpenRouter	1	50	50	99.77%	\$0.11453020	15550.28 ms	15376.22 ms
OpenRouter	2	50	50	99.78%	\$0.09520547	11441.53 ms	10863.19 ms
OpenRouter	3	50	50	99.78%	\$0.09386227	13438.79 ms	13273.32 ms
OpenRouter	4	50	50	95.03%	\$0.10529059	15630.55 ms	14865.37 ms

Price First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	99.88%	\$0.01706100	14751.31 ms	14009.91 ms
Infron	2	50	50	99.89%	\$0.00571700	18265.77 ms	18100.26 ms
Infron	3	50	50	99.89%	\$0.00566500	13939.09 ms	13719.39 ms
Infron	4	50	50	99.88%	\$0.00549600	13603.89 ms	12556.91 ms
OpenRouter	1	50	50	99.54%	\$0.12981646	13027.87 ms	13007.17 ms
OpenRouter	2	50	50	99.78%	\$0.09520547	12565.22 ms	12426.92 ms
OpenRouter	3	50	50	97.50%	\$0.09241668	11934.01 ms	11485.97 ms
OpenRouter	4	50	50	99.78%	\$0.08432161	8656.02 ms	8579.21 ms

Latency First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	99.72%	\$0.17094900	6315.97 ms	5713.48 ms
Infron	2	50	50	99.73%	\$0.14970000	5585.05 ms	5425.40 ms
Infron	3	50	50	99.73%	\$0.14835600	5339.92 ms	4755.13 ms
Infron	4	50	50	99.72%	\$0.14343400	6386.33 ms	5049.11 ms
OpenRouter	1	50	50	99.77%	\$0.08894083	5777.97 ms	5635.33 ms
OpenRouter	2	50	50	99.78%	\$0.09520547	5771.35 ms	5327.51 ms
OpenRouter	3	50	50	99.77%	\$0.09707411	5875.42 ms	5644.30 ms
OpenRouter	4	50	50	99.77%	\$0.08894083	6308.01 ms	5730.93 ms

TTFT First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	99.72%	\$0.14343400	6669.23 ms	5187.90 ms
Infron	2	50	50	99.73%	\$0.14970000	5649.02 ms	4967.96 ms
Infron	3	50	50	99.73%	\$0.14835600	5246.79 ms	4909.10 ms

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	4	50	50	99.72%	\$0.14343400	5065.79 ms	4639.46 ms
OpenRouter	1	50	50	99.76%	\$0.10181368	6471.43 ms	5376.22 ms
OpenRouter	2	50	50	99.76%	\$0.12092926	6007.83 ms	5451.96 ms
OpenRouter	3	50	50	99.76%	\$0.11958606	6069.91 ms	6045.35 ms
OpenRouter	4	50	50	99.75%	\$0.11466462	5488.56 ms	5183.79 ms

10. Discussion: Business Value, Boundaries, and Engineering Implications

Business decisions should not rely on one metric. Stable long-context and high-frequency template workloads should prioritize cache rate and cost; realtime interaction must constrain TTFT and E2E latency; batch processing often prioritizes throughput and failure cost.

Routing mode	Business objective	Observed result	Scenarios	Caveat
Throughput First	Maximize output capacity per unit time	Infron leads overall (3/5 metrics)	Batch generation, offline summaries, backend processing	Decide with budget, SLA, and cache stability together
Price First	Minimize request and token cost	OpenRouter leads overall (3/5 metrics)	High-frequency templates, support automation, marketing, RAG prefixes	Cache and cost are stronger; still check speed SLA
Latency First	Minimize full-response waiting time	Infron leads overall (3/5 metrics)	Online chat, agent chains, IDE/writing assistants, realtime tools	Decide with budget, SLA, and cache stability together
TTFT First	Minimize streaming first-token time	Infron leads overall (3/5 metrics)	Streaming chat, realtime copilots, first-screen feedback	Decide with budget, SLA, and cache stability together

11. Conclusion

Routing-Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	OpenRouter	OpenRouter	Infron	Infron	Infron	Infron leads overall (3/5 metrics)
Price First	Minimize request and token cost	Infron	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Latency First	Minimize full-response waiting time	OpenRouter	OpenRouter	Infron	Infron	Infron	Infron leads overall (3/5 metrics)
TTFT First	Minimize streaming first-token time	OpenRouter	OpenRouter	Infron	Infron	Infron	Infron leads overall (3/5 metrics)

12. Limitations, Missing Data, and Next Experiments

Limitation	Impact	Next step	Current handling
Full routing trace	Cannot prove every provider choice, fallback, and retry path hop by hop	Add provider routing trace, decision logs, and fallback reasons	Use only returned provider fields and provider distribution
Longer time window	4x50 observes short-window stability but not day-level drift	Add soak tests and repeated windows	Scope conclusions to this run
Production corpus	Built-in templates do not cover every workload distribution	Use sanitized production-stratified corpora	Discuss representative long-context templates only
Cost-field consistency	Cost coverage and semantics may differ by platform	Reconcile with billing and provider cost breakdown	Use only explicitly returned cost fields

13. Reproducibility Appendix

Artifact	Path
Summary	export/mimo25_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions summary.json
Paired dataset	export/mimo25_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions benchmark_pairs.csv
Request-level dataset	export/mimo25_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions benchmark_requests.jsonl
Filtered structured records	export/mimo25_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions records.json
Excluded-record audit	export/mimo25_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions records_excluded.json
Test source	tests/test_rerun_routing_sort_cache_cost_ab.py

Artifact	Path
Benchmark runner source	<code>scripts/rerun_routing_sort_cache_cost_ab.py</code>
HTML report renderer source	<code>scripts/render_glm52_deepseek_style_report.py</code>
Dataset reference	<code>business_representative</code> built-in representative business templates; request-level export is <code>benchmark_requests</code>